# Discussion of “Job Transformation, Specialization, and the Labor Market Effects of AI” by Lukas Freund and Lukas Mann

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# Motivation and Background

- *The* question of our time: What will be the impact of AI on
  - Pedagogy,
  - Innovation,
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- Range of outcomes is immense

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Past literature on AI and work practices has:

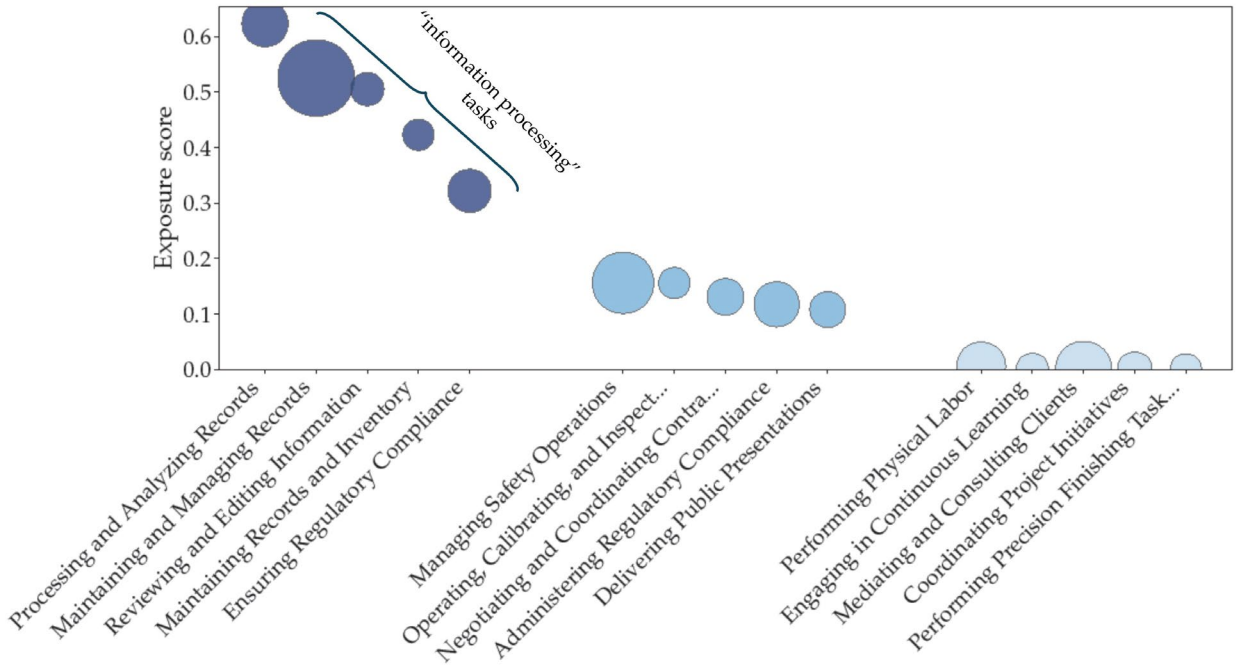
- Studied earlier implementations of AI (Brynjolfsson, Li, Raymond 2023; Hampole et al, 2025; Humlum and Vestergaard 2025); or
- Used LLM to assess which occupations/tasks are exposed to AI in the medium-to-long run (Eloundou et al. 2023)

Paper's Question: How will AI reshape job tasks, occupational sorting, and inequality?

# What Does This Paper Do?

1. Models occupations as bundles of tasks
   - Workers have multidimensional skills for producing these tasks. Choose where to work and how much time to allocate to each task according to their skills.
   - AI changes (a) the weights associated with each task and (b) occupation-specific productivity.
2. Constructs occupation-task weights from O*NET and LLM
   - Groups detailed O*NET Detailed Work Activities into 38 tasks.
   - Asks LLM: What fraction of time in occupation o is taken by task t before/after 2000.
3. Estimates distribution of worker skills using NLSY97
   - If we see worker i in occupation o (or o'), it must be because they must be relatively skilled in tasks important to o (or o')
   - Wages help identify i's absolute skill levels.
4. Considers occupation-task pairs' exposure to AI from Eloundou et al. (2023, "GPTs are GPTs")

# How do Workers Respond to AI Automation?




Paper draws on Eloundou et al. for tasks' exposure to AI

LLM's estimates of the fraction of time spent in different tasks by occupation will also be important.

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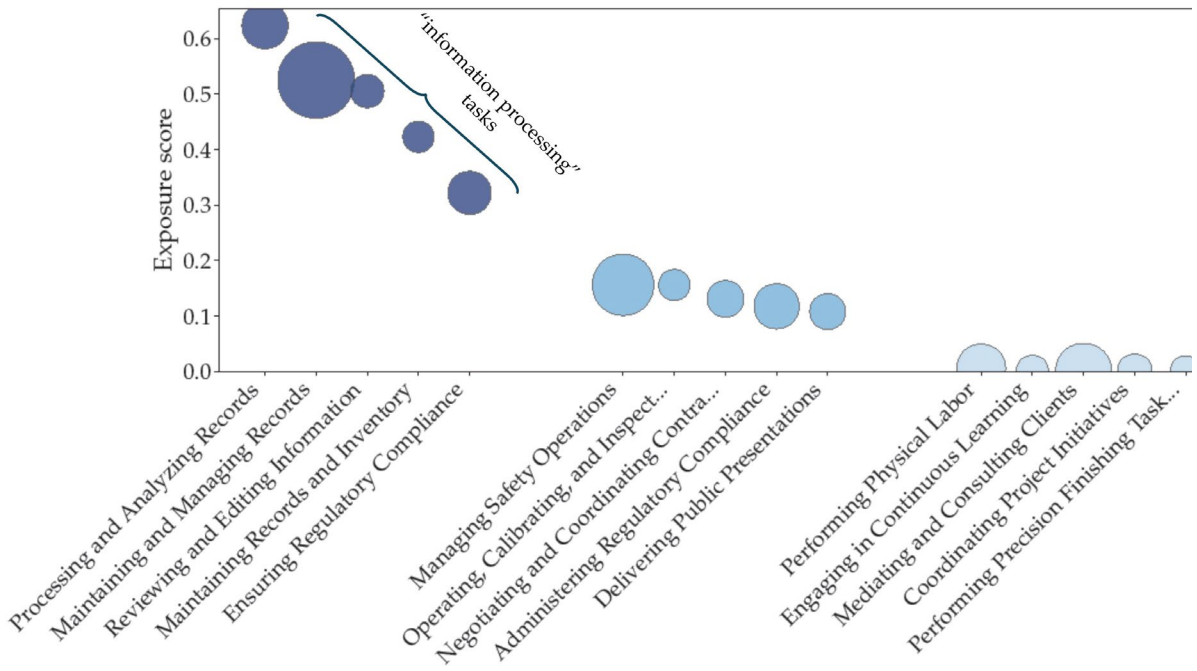


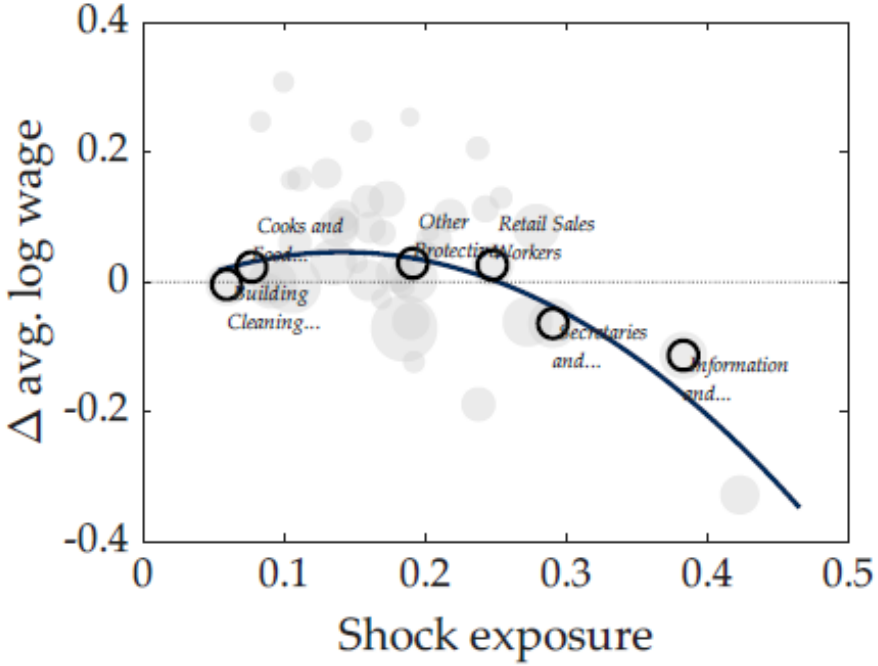


Most exposed occupations include those in Office and Administrative Support.

Workers in these occupations see their wages drop by 10+ percent.

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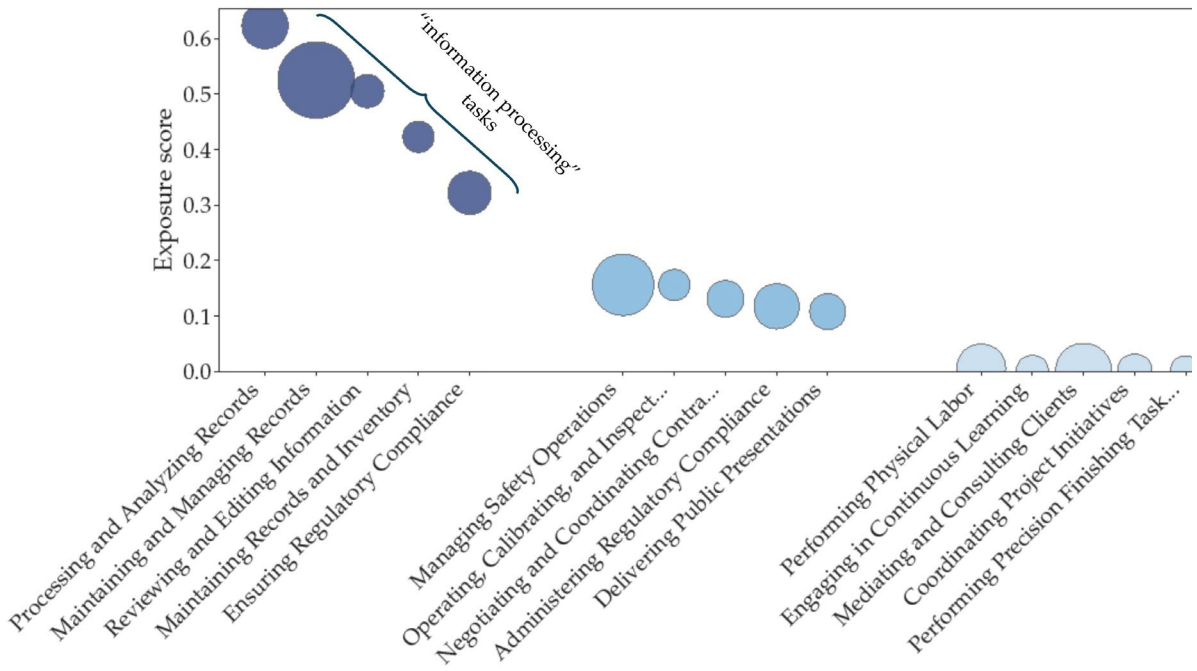
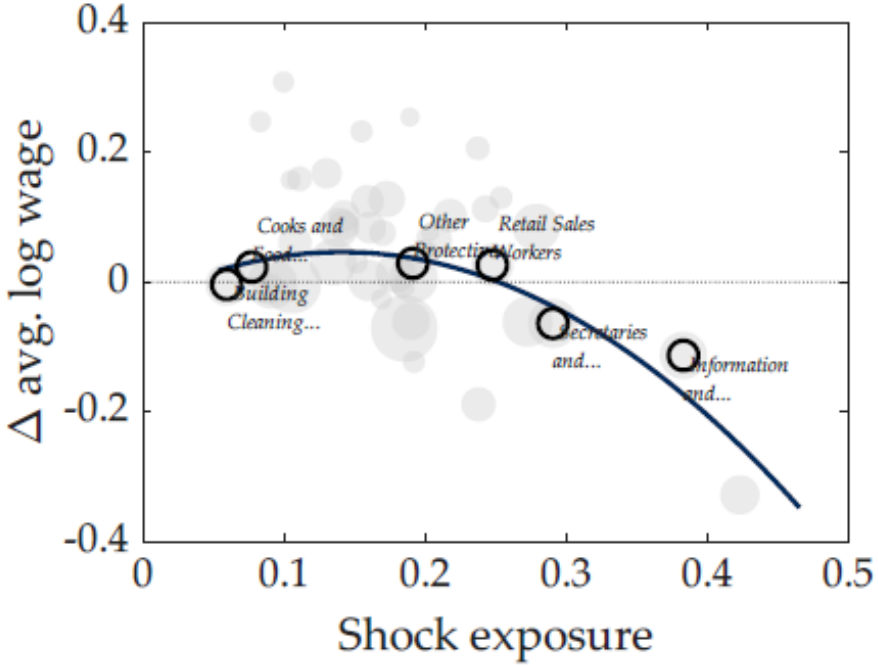
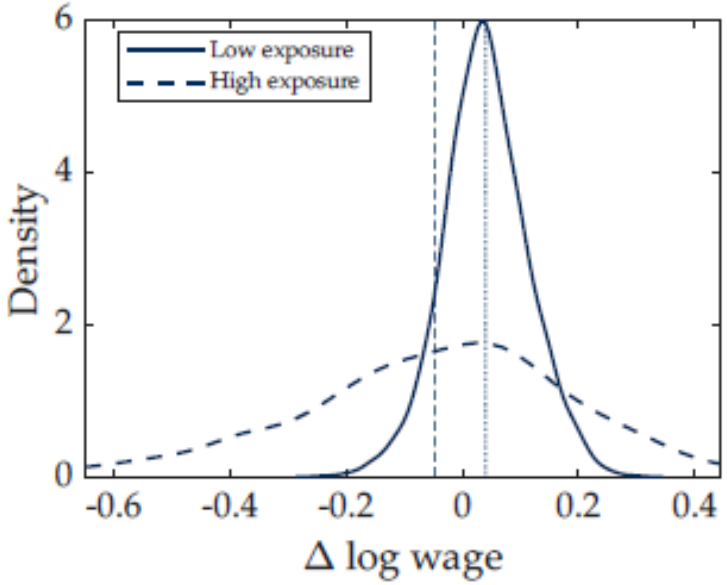



Within occupations, some workers may benefit from AI exposure, others may lose. Depends on:

- whether the exposed tasks are those I have a comparative advantage in;
- whether the other skills I have are demanded in my occupation



# A New Paradigm for Measurement?

- *Silicon sampling*: Artificial participants stand in for real people.
- Benefits
  - Dramatic increase in detail and access at low cost.
  - Across multiple contexts, LLMs provide realistic measures of human behavior, similar to what humans would themselves provide (Brand, Israeli, Ngwe, 2025; O*NET, 2026).
  - LLMs will only get more powerful from here.
- Caveats
  - Hallucinations, while becoming less obvious and prevalent over time, will continue.
  - The data generating process becomes a black box.
    - Need new methods to think about sampling variability, etc.
- Rossi, Harrison, Shklovski (2024, “The Problems of LLM-Generated Data in Social Science Research”) have an even-handed discussion.

# Which Tasks Are Exposed to AI?

- Question: How has ChatGPT's own assessments of occupational exposure changed over the last 2.5 years?

- Method: Re-do prompting of Eloundou et al. with newer model.

- Result: Exposure indices lower, more concentrated in Office and Administrative Support and Computer Occupations.

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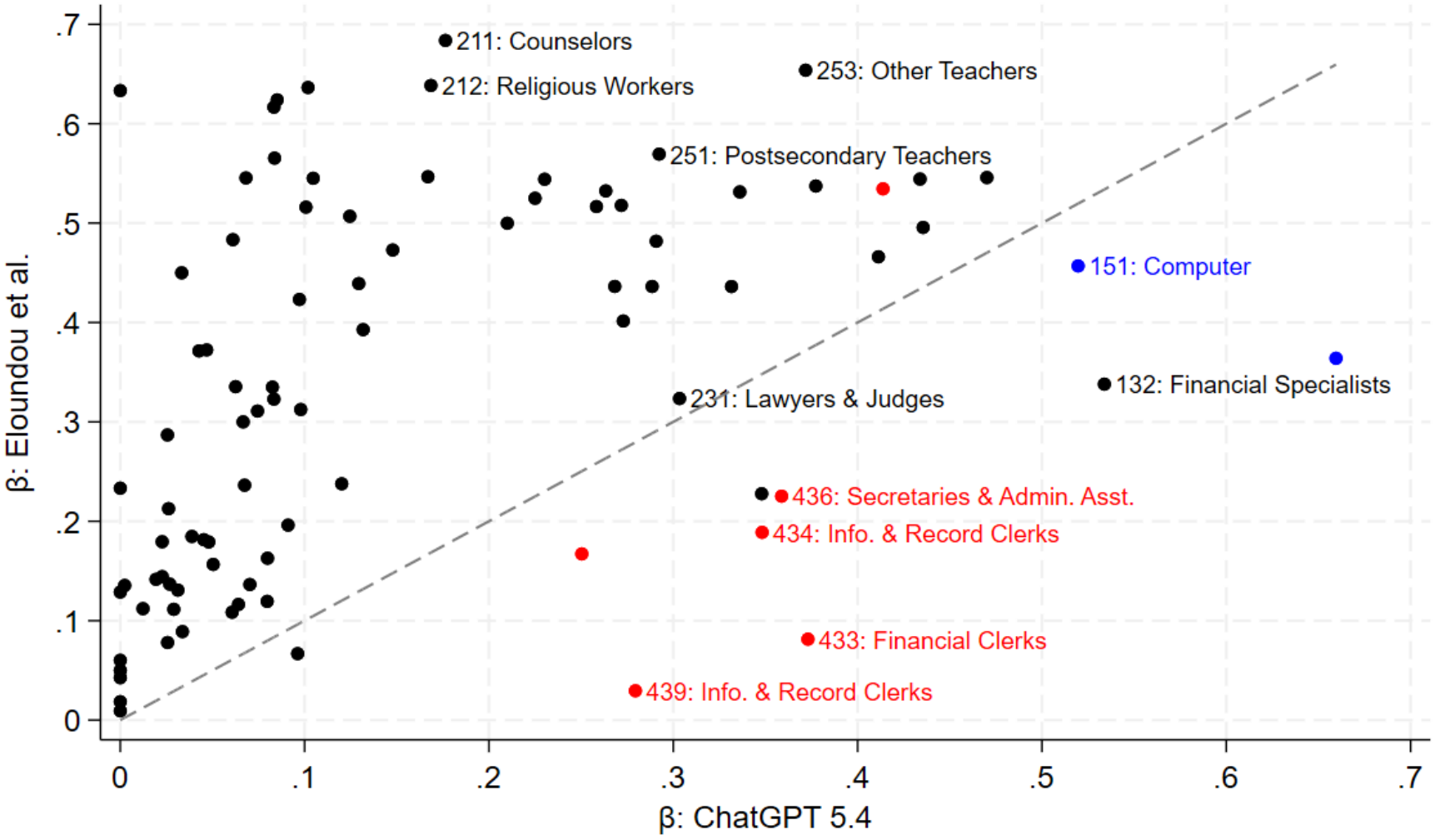

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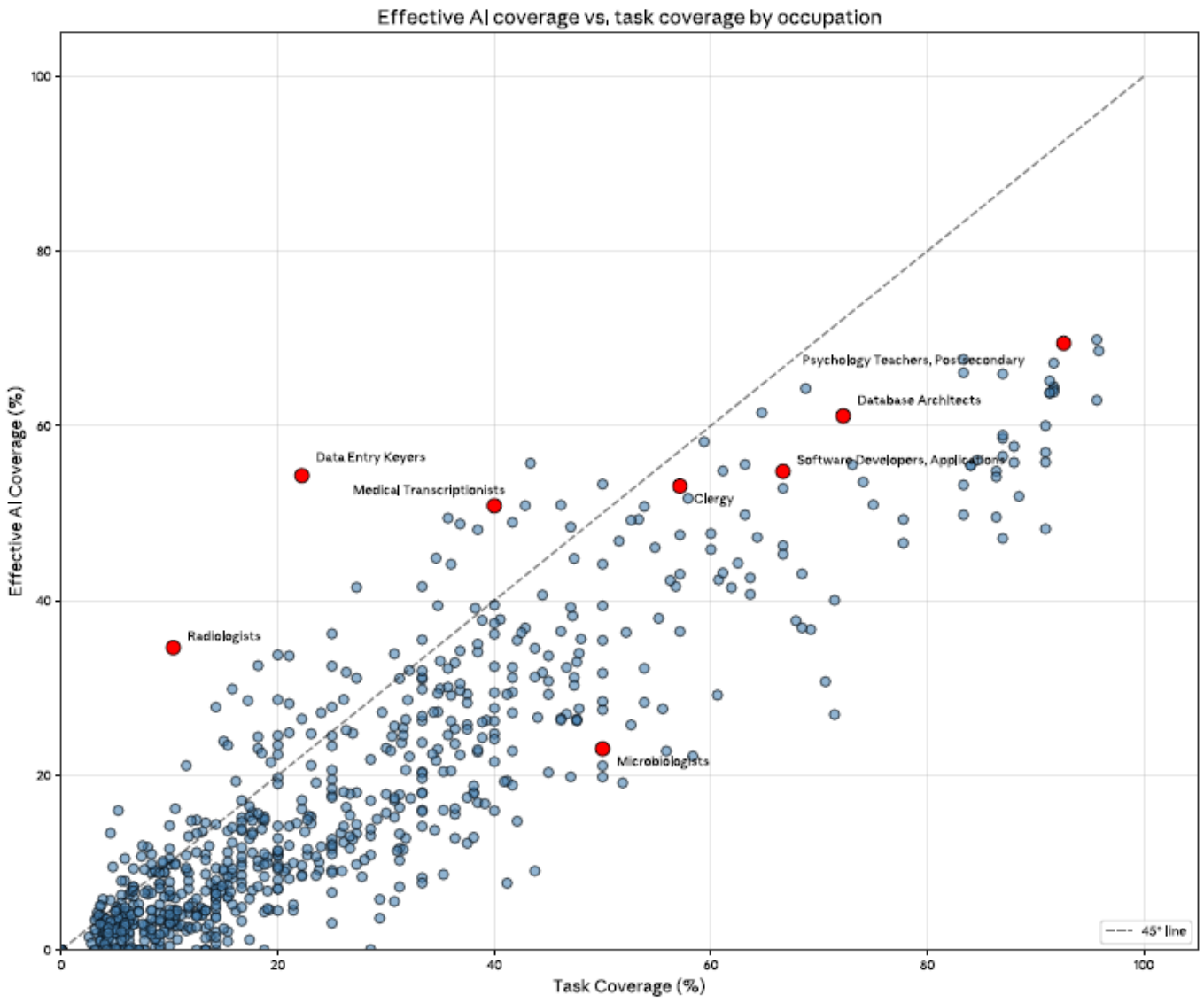
- Anthropic publishes a database of tasks' exposure to Claude as of fall 2025. (Appel et al., 2026, "The Anthropic Economic Index Report: Economic Primitives")
- Distinction between raw task exposure (horizontal axis) and task exposure accounting for AI's failure at performing certain tasks (vertical axis)